

Dynamic Graph Convolutional Network-based Long Short-Term Memory for Taxi Trip Time Prediction

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Abstract—In the last few years, traveling time has increased rapidly due to improper management of the time schedule and road traffic which also directly affects taxi users. The existing models do not offer appropriate results for the prediction of time taken to complete a taxi trip due to poor spatial and temporal information of the features. This paper proposes a Dynamic Graph Convolutional Network-Long Short-Term Memory (DyGCN-LSTM) model for the prediction of taxi trip time. The Extreme Gradient Boosting (XGB) model method is used for the analysis of feature importance based on Shapley Additive Explanations (SHAP) value. Mean Absolute Error (MAE), precision, Root Mean Squared Error (RMSE), recall, and f1-score are the metrics taken into consideration for performance evaluation. The DyGCN-LSTM is prone to a low error rate of 7.05 for RMSE and 2.04 for MAE on the New York City (NYC) taxi trip record-2019 dataset, with minimum error of 5.22 for RMSE and 1.65 for MAE on the NYC taxi trip record-2022 dataset. This has hence proved to be superior in comparison to the existing Long Short-Term Memory - Convolutional Neural Network (LSTM-CNN) autoencoder method.

Keywords—dynamic graph convolutional network (DYGCN), extreme gradient boosting (XGB), long short-term memory (LSTM), new york city (NYC), prediction, taxi trip.

I. INTRODUCTION

The whole world is dependent on taxicab services to reach destinations at a shorter range of time [1]. The taxi fare usually depends on the distance travelled for the particular time spent where routes have fixed rates for the destination [2]. The government provides license to drivers only if the taxis meet their safety standards. As the technology is progressing, Global Positioning System (GPS) navigation is installed in taxis to find out an efficient route for traveling [3]. GPS also helps monitor and track taxis for safety purposes. Also, various mobile applications such as Ola and Uber have been introduced to book rides so that passengers can easily get a taxi. The urban development system is completely dependent on taxi services due to its flexibility [4]. However, the prediction of the taxi travel time is difficult due to heavy traffic and signals on the road that affect the efficiency and performance of taxi services [5]. The DL methods are used to increase the prediction rate for taxi travel time so that users can plan their trip based on the traveling time. The contributions of this work are summarised as follows:

- The XGB is used to identify the importance of features based on the SHAP values for gathering significant information of data based on the analysis of all the possibilities of feature interaction.
- The DyGCN captures approximate spatial information of the features based on the network topology of the location coordinates to improve the performance of prediction.
- The DyGCN-LSTM captures both spatial and temporal information related to traffic and locations in the networks, which further increases the prediction of taxi trip time.

The remaining sections of the paper are structured as follows: Section 2 presents the literature review, Section 3 details the proposed methodology, Section 4 discusses the experiments and results, while Section 5 displays the discussion and finally, Section 6 states the conclusion of this study.

II. LITERATURE REVIEW

Mai et al. [6] presented a Geographical and Temporally Weighted Regression (GTWR) model to determine the traveling demand for taxis during COVID-19. The Ordinary Least square (OLS) method was used to analyze the variable relationship using global and local regression models. The GTWR model helped to measure the values for temporal and spatial to identify the exact taxi travel demand that enhanced the performance of the model. However, the model failed to identify the specified features due to poor learning ability which minimized the model's efficiency.

Ghandeharioun and Kouvelas [7] suggested a taxi traveling time estimation model for arterial networks using GPS data. The allocation of the travel time was adjusted by capturing random variables. The important details of the data were selected based on the spatial correlation of the features that improved the estimation of taxi travel time. However, the model's performance was affected due to the lack of optimization technique of the model that caused overfitting issues and reduced the accuracy of the model.

Elsir et al. [8] introduced a unified DL mechanism using the Joint Spatial-Temporal Correlation (JSTC) model to

predict taxi travel time. The temporal convolutional network (TCN) captured spatial regions of the data based on temporal dependencies. The Rectified Linear Unit (ReLU) attention mechanism was used to get the significant details of features which increased the prediction accuracy of taxi travel time. However, the model was limited due to a lack of proper details of features related to the factor of taxi travel that minimized the prediction rate.

Ghandeharioun et al. [9] developed a short-term passenger demand prediction hybrid model to forecast traffic. LSTM-CNN classifier was used to predict the future demand for taxi travel. The LSTM acted as an encoder and CNN acted as a decoder to get the important details of the feature that advanced the prediction rate of the model to understand the taxi demand. However, the performance of the model was affected due to the noises generated during the training of the model which led to misclassification of data and decreased prediction accuracy.

Rathore et al. [10] implemented a taxicabs price prediction model based on a hybrid clustering model. The Ordinal Logistic Regression (OLR) model was used as a classifier to predict the taxicab price. The features were clustered based on the timestamps. The variable importance technique was applied to get significant information from the data that increased the accuracy of taxi price prediction. However, the model did not capture the underlying features due to the lack of identifying the edge data points that decreased the accuracy of taxi price prediction.

Tygesen et al. [11] presented a Graph Neural Network (GNN) model to predict the transportation time for the taxi ride. The Graph Network (GN) blocks were used to encode global features present in the data. The Gated Recurrent Unit (GRU) model was used as a decoder to get the temporal dependencies. The GNN analyzed the features that enhanced the prediction of taxi transportation time. However, during the training process, some of the gradients vanished that led to suboptimal performance hence reduction in the model's performance.

Dokuz [12] suggested a taxi trajectory model for the estimation of regional traffic based on temporal and special information. The Regional Traffic Velocity Estimation (RTVE) model was used to estimate the taxi trajectory. Based on the special and temporal features, the RTVE calculated the shortest distance for the taxi trip which advanced the estimation of the traffic accurately. However, the model failed to interpret the significant feature of the data due to the dimensionalities of the features that minimized the efficiency of taxi trajectory estimation.

Tao et al. [13] introduced a Spatial-Temporal Attention-based Graph Convolution Network (MISTAGCN) for the prediction of traffic. The Variational Auto-Encoder (VAE) and GCN methods were used to get the information of the traffic. The temporal convolution and residual connection were used to handle the dimensionalities of the features that also advanced the model's performance. However, improper selection of features created overfitting issues that affected the model's performance.

III. PROPOSED METHODOLOGY

The NYC taxi trip records of the 2019 and 2022 datasets are considered for the prediction of taxi trip time. The min-max scalar technique is used to normalize the ranges of data.

The XGB model is used to get the important features based on the SHAP values. The proposed DyGCN-LSTM model captures the spatial and temporal information of the features to predict the taxi trip time. Figure 1 represents the block diagram of the proposed methodology.

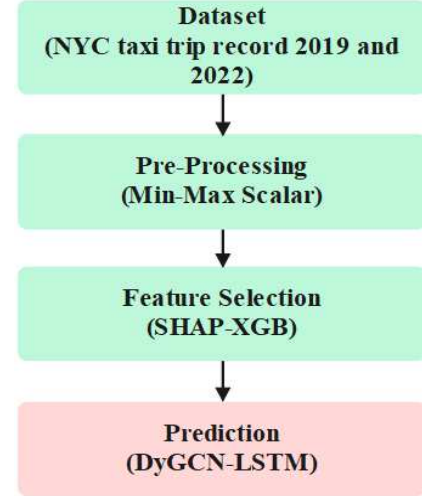


Fig. 1. Block diagram of the proposed methodology

A. Dataset

The NYC taxi trip record data for 2019 and 2022 are publicly available datasets considered for the prediction of taxi trip time [14][15]. The data was obtained from the NYC Taxi and Limousine Commission (TLC). The NYC taxi trip record-2019 dataset consists of 3.94 million samples of taxi trips with 18 different features and the NYC taxi trip record-2022 dataset consists of 2.4 million samples of taxi trips with 19 different features. The data mainly consists of Vender ID, trip pickup data time, trip drop-off date time, passenger count, rate code ID, pickup location ID, trip distance, drop-off location ID, store, and forward flag, and payment time details. The NYC taxi data is input into the min-max scalar where the data gets normalized for the next process.

B. Pre-Processing

The NYC taxi data is input for pre-processing where different ranges of the data are normalized to avoid biases [9]. The min-max scalar technique is used to normalize the data to get the fixed range values [0,1] using eq. (1).

$$X_{normalised} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

Where, $X_{normalised}$ and X are respectively the normalized and actual values of the features, X_{max} and X_{min} are the maximum and minimum values of features. The Extrema Gradient Boosting (XGB) model is used to evaluate the importance of features to select the significant details of the taxi data.

C. Feature Selection

The XGB is used to find the importance of the features based on the SHAP values to gather the significant details of the data. The feature importance is selected by calculating the possibilities of all interactions of the features using eq. (2).

$$Importance_j = \frac{1}{N} \sum_{i=1}^N |\phi_{ij}| \quad (2)$$

Where N is the number of samples used for the analysis of the model's interpretability ϕ_{ij} is the SHAP value of a feature j for instance i . The XGB increases the interpretability of the model to enhance the predictive performance. The DyGCN-LSTM classifiers predict the time for taxi trip based on the importance of features. Based on the SHAP values and XGB importance, the features analysis of 10 features is processed to predict the taxi trip time. Trip pickup data time, trip distance, trip drop-off date time, pickup latitude, pickup longitude, drop-off longitude, drop-off latitude, toll amount, fare amount, and extra are the features selected from the data.

D. Prediction

The DyGCN-LSTM classifier is proposed for the prediction of taxi trip time based on the importance of features. The DyGCN captures significant spatial information of pickup time, drop off time, drop off location, pickup location, traffic, and road networks from the data. The captured spatial information is forwarded to the LSTM model to get the temporal patterns of the data for traffic of day, week, or month. Based on the spatial and temporal information of the data, the model predicts the taxi trip time. The DyGCN uses the temporal graph distance correlation on the basis of adjacent matrix to encode the topology structure of the road network and road attributes using eq. (3).

$$f(X, A_{Dis_{i,j}}^{corr}) = \sigma(\hat{A}_{Dis_{i,j}}^{corr} Relu(\hat{A}_{Dis_{i,j}}^{corr} \times X \times W_0) W_1) \quad (3)$$

Where, X and $A_{Dis_{i,j}}^{corr}$ represents the feature matrix and distance correlation adjacent matrix $\hat{A}_{Dis_{i,j}}^{corr} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$, where, $\tilde{A}$ denotes the self-connected matrix and $\tilde{D}$ is the diagonal degree matrix. The W_0 and W_1 are the learnable weight matrix, σ and $Relu$ are the sigmoid and ReLU activation functions. The output of $f(X, A_{Dis_{i,j}}^{corr})$ is used by the DyGCN-LSTM model. The linear operations of the LSTM model are replaced by the DyGCN module given in eq. (3). The modification of the proposed DyGCN-LSTM model is given from eq. (4-9).

$$F_g^t = \sigma(W_{if}[f(A_{corr}^{Dis}, X_t), S_{t-1}] + B_{if}) \quad (4)$$

$$I_g^t = \sigma(W_{ii}[f(A_{corr}^{Dis}, X_t), S_{t-1}] + B_{ii}) \quad (5)$$

$$\tilde{C}_t = \tanh(W_{ic}[f(A_{corr}^{Dis}, X_t), S_{t-1}] + B_{ic}) \quad (6)$$

$$C_t = F_g^t \times C_{t-1} + I_g^t \times \tilde{C}_t \quad (7)$$

$$O_t = \sigma(W_{io}[f(A_{corr}^{Dis}, X_t), S_{t-1}] + B_{io}) \quad (8)$$

$$S_t = O_t \times \tanh(C_t) \quad (9)$$

Where, F_g^t is the modified forget gate of the LSTM model, I_g^t is input gate, $\tilde{C}_t$ is candidate state, C_t is cell state, O_t is output state at time t and S_t is the hidden state of the output. B_{if}, B_{ii}, B_{ic} and B_{io} terms are the dimensions and W_{if}, W_{ii}, W_{ic} and W_{io} are the weights of the features. The DyGCN is used to improve the performance of the LSTM model for non-linear operations by capturing the spatial and temporal information of the features to predict the approximate taxi trip time.

IV. EXPERIMENT AND RESULTS

The DyGCN-LSTM performance is executed in Python software and the configuration requirements of the system is Windows 10, i7 processor, and 16 GB RAM. The RMSE, MAE, precision, recall, and f1-score are the metrics considered for the evaluation of DyGCN-LSTM which is shown in eq. (10-14).

$$RMSE = \sqrt{\frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |X_{t,i} - \hat{Y}_{t,i}|^2} \quad (10)$$

$$MAE = \frac{1}{N \times T_p} \sum_{t=1}^{T_p} \sum_{i=1}^N |X_{t,i} - \hat{Y}_{t,i}| \quad (11)$$

$$Precision = \frac{TP}{FN + TP} \quad (12)$$

$$Recall = \frac{TP}{FP + TP} \quad (13)$$

$$F1 - Score = \frac{2(Precision \times Recall)}{Precision + Recall} \quad (14)$$

Where, $X_{t,i}$ and $\hat{Y}_{t,i}$ are real and predicted values, T_p is the traffic behavior for the time t , N represents the nodes of the network, TP is the true positive value, TN is the true negative value, FP is the false positive value and FN is the false negative value.

A. Performance of DyGCN-LSTM

The DyGCN-LSTM performance is analyzed using two datasets namely NYC taxi trip record of the 2019 and 2022 datasets in terms of f1-score, MAE, RMSE, precision, and recall metrics. Table 1 and Table 2 represent the performance of DyGCN-LSTM for the NYC taxi trip record 2019 and 2022 dataset.

TABLE I. PERFORMANCE OF DYGCN-LSTM FOR NYC TAXI TRIP RECORD-2019 DATASET

Method	RMSE	MAE	Precision	Recall	F1-Score
RLSTM	10.58	5.36	0.591	0.569	0.579
FCN-LSTM	9.63	4.06	0.789	0.692	0.737
RNN-LSTM	8.20	3.79	0.896	0.759	0.821
DyGCN-LSTM	7.05	2.04	0.948	0.827	0.883

From Table 1, the DyGCN-LSTM is seen to exhibit better performance in terms of f1-score, MAE, RMSE, precision, and recall for NYC taxi trip record 2019 dataset. The DyGCN-LSTM shows less error values of 7.05 RMSE and 2.04 MAE with improved precision of 0.948, f1-score of 0.883, and recall of 0.827.

TABLE II. PERFORMANCE OF DYGCN-LSTM FOR NYC TAXI TRIP RECORD-2022 DATASET

Method	RMSE	MAE	Precision	Recall	F1-Score
RLSTM	8.95	4.93	0.456	0.483	0.469
FCN-LSTM	7.27	3.59	0.717	0.611	0.659
RNN-LSTM	6.42	2.78	0.825	0.741	0.780
DyGCN-LSTM	5.22	1.65	0.953	0.855	0.901

From Table 2, the DyGCN-LSTM presents an improved performance in terms of precision, f1-score, RMSE, recall, and MAE for the NYC taxi trip record 2022 dataset. The DyGCN-LSTM displays less error values of 5.22 RMSE and

1.65 MAE, with an enhanced precision of 0.953, f1-score of 0.901, and recall of 0.855.

B. Comparative Analysis

TABLE III. COMPARATIVE ANALYSIS

Dataset	Method	RMSE	MAE	Precision	Recall	F1-Score
NYC taxi trip record-2019	LSTM-CNN Autoencoder [9]	8.28	NA	NA	NA	NA
	DyGCN-LSTM	7.05	2.04	0.948	0.827	0.883
NYC taxi trip record-2022	OLR [10]	NA	NA	0.900	0.774	0.832
	MISTAGCN [13]	6.27	2.18	NA	NA	NA
	DyGCN-LSTM	5.22	1.65	0.953	0.855	0.901

The comparative of proposed method is represented in this section with the existing metrics in terms of recall, RMSE, MAE, precision, recall, and f1-score on the NYC taxi trip record 2019 and 2022 datasets. The LSTM-CNN Autoencoder [9], OLR [10], and MISTAGCN [13] performances are compared with the proposed method. Table 3 shows the comparative analysis.

V. DISCUSSION

In this section, the drawbacks of the existing methods and the advantages of the proposed DyGCN-LSTM are discussed. The existing methods LSTM-CNN Autoencoder [9] did not achieve the expected prediction value due to the noises generated during the model's training. The OLR [10] failed to identify the edge data points that minimized the performance. The MISTAGCN [13] performance was affected due to improper selection of features that increased the error rate. The proposed DyGCN-LSTM captured both spatial and dynamic information of the data based on the traffic network topology that improved the taxi trip time prediction. The DyGCN-LSTM has obtained minimum error values of 7.05 for RMSE and 2.04 for MAE and 5.22 for RMSE and 1.65 for MAE respectively on the NYC taxi trip record 2019 and 2022 datasets.

VI. CONCLUSION

This paper proposes a DyGCN-LSTM for the prediction of taxi trip time by capturing the spatial and temporal information of the traffic networks from the data. The DyGCN captures the spatial details of pick-up and drop-off locations using non-linear features and fed into the LSTM to capture the temporal patterns of the data based on the traffic details for the particular instance which further elevates the performance of the model to predict the taxi trip time. The f1-score, RMSE, precision, MAE, and recall are the metrics considered for the evaluation of DyGCN-LSTM performance. The proposed DyGCN-LSTM shows lesser error values of 7.05 for RMSE and 2.04 for MAE and 5.22 for RMSE and 1.65 for MAE with NYC taxi trip record 2019 and 2022 datasets, respectively. In the future, different optimization-based DL algorithms can be incorporated to enhance the prediction of taxi trip time.

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